

# Renewable Energy Generation Analysis Across Indian States

## 1. Project Overview and Objective

This project focuses on analysing Indian renewable energy generation data from 2020 to 2024 using Python and Power BI. The dataset contains state-wise daily generation data for solar, wind, and other renewable energy sources, along with capacity and total generation including renewable energy. The analysis identifies trends, seasonal patterns, top and bottom-performing states, and changes in renewable energy generation over time. It also compares the contribution of different renewable sources to understand their impact on overall renewable generation.

## 2. Data Sources

- **Source Description and Timeline:** the daily renewable energy dataset taken from GitHub.
- **Period of Dataset:**2020-2024
- **Domain:** Energy and Power Generation

## 3. Problem Statement

Renewable energy generation varies significantly across Indian states and over different time periods. There is a need to analyse these variations and understand the contribution of solar, wind, and other renewable sources. This project aims to identify key trends, performance differences, and seasonal patterns in renewable energy generation from 2020 to 2024. The analysis helps provide data-driven insights into renewable energy performance across India.

## 4. Attribute (Column /Features) Details:

| Attribute Name | Data Type | Description                                                                      |
|----------------|-----------|----------------------------------------------------------------------------------|
| Date           | Date      | The date on which power generation was recorded                                  |
| Region         | Text      | The geographical region                                                          |
| State name     | Text      | State name in India                                                              |
| Gen Actual     | Decimal   | The actual electricity generated during the day, measured in Million Units (MU). |
| Gen Programmed | Decimal   | The planned electricity generation for the day, measured in Million Units (MU).  |
| Capacity       | Decimal   | The power generation capacity, measured in Megawatts (MW).                       |
| Wind Energy    | Decimal   | The electricity generated from wind energy, measured in Million Units (MU).      |
| Solar Energy   | Decimal   | The electricity generated from solar energy, measured in Million Units (MU).     |

|                        |         |                                                                                                                           |
|------------------------|---------|---------------------------------------------------------------------------------------------------------------------------|
| Other renewable energy | Decimal | The electricity generated from other renewable energy sources measured in Million Units (MU).                             |
| Total renewable energy | Decimal | The combined electricity generation from wind, solar, and other renewable energy sources, measured in Million Units (MU). |
| Total Gen with Re      | Decimal | The total electricity generation including renewable energy generation, measured in Million Units (MU).                   |

---

## 5. Tools & Technologies

### Python:

- Pandas – Data Cleaning & Data Transformation
- Seaborn – Statistical Data Visualization
- Matplotlib – Data Visualization
- Data Type Handling
- Missing Value Treatment
- Duplicate Detection and Removal
- Exploratory Data Analysis (EDA)

### Power BI:

- Data Import and Transformation
- DAX Measures
- KPI Creation
- Interactive Dashboard Creation
- Trend and Comparative Analysis
- Data Visualization

## 6. Data Pre-Processing (Python)

### Data Cleaning:

- Checked and corrected data types
- Removed duplicate records
- Identified and treated missing values
- Checked inconsistent state and region names
- Identified negative values in renewable energy columns
- Verified the accuracy of total renewable energy values

### Data Transformation:

- Converted the Date column into proper date format
- Created Year and Month columns for time-based analysis
- Calculated Total Renewable Energy from solar, wind, and other renewable sources
- Filtered the dataset for the required analysis period (2020–2024)
- Prepared data for further analysis and visualization

### Exploratory Data Analysis:

- Analysed state-wise and region-wise generation patterns
- Compared solar, wind, and other renewable energy generation
- Identified top and bottom-performing states
- Analysed yearly and monthly generation trends
- Studied variations and patterns using statistical summaries

### Data Visualization:

- Created bar charts for state and region comparisons
- Created histograms to understand energy distribution
- Created scatter plots to analyse relationships between variables
- Created trend and comparison charts using Seaborn and Matplotlib

## 7. Data Modelling and DAX (Power BI)

### Data Model:

Prepared the cleaned renewable energy dataset and organized the fields for effective analysis and visualization in Power BI.

### DAX Measures:

- Total Renewable Energy
- Total Solar Energy
- Total Wind Energy
- Renewable Energy Contribution %

---

## 8. Analysis and Visualizations (Power BI)

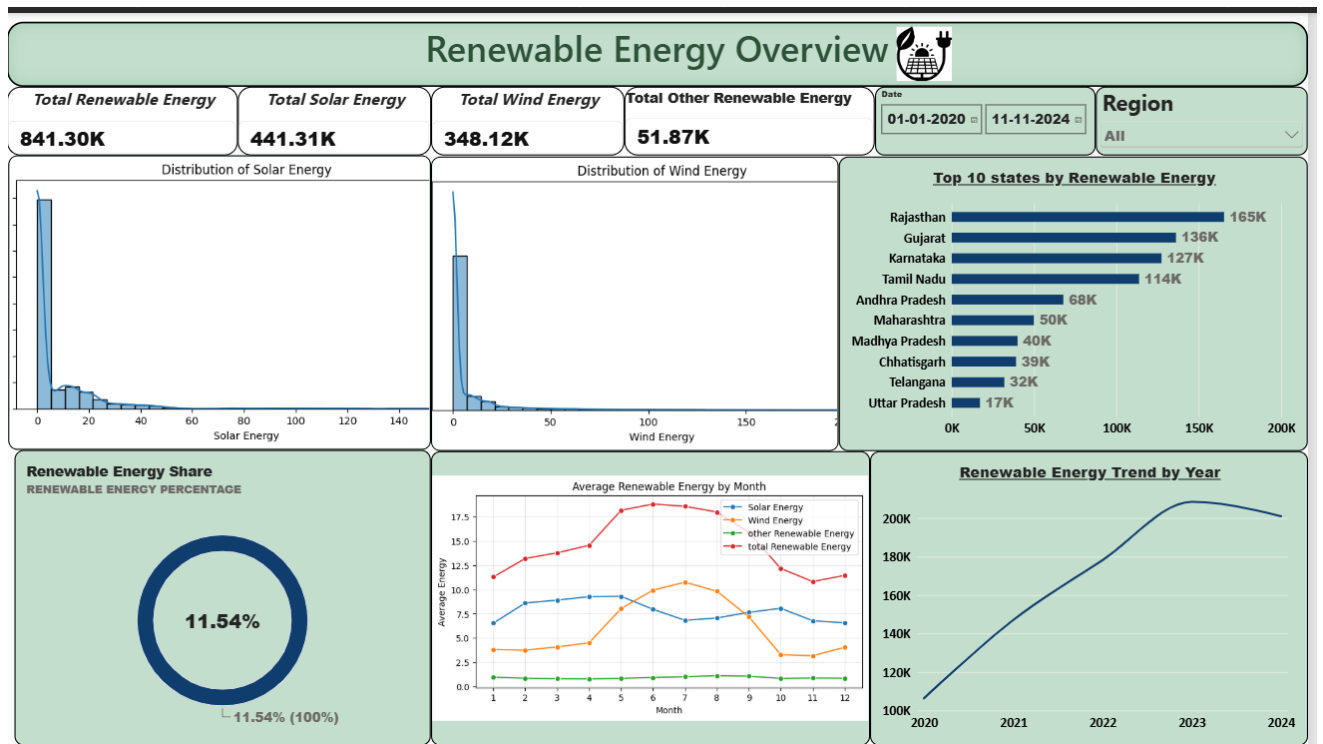
### Dashboard 1: Renewable Energy Overview

#### KPI Cards

- Total Renewable Energy
- Total solar Energy
- Total Wind Energy
- Total Other Renewable Energy

#### Charts

- Solar Energy distribution (Histogram)
- Wind Energy distribution (Histogram)
- Top 10 states by Renewable Energy (Bar chart)
- Renewable Energy share (Donut Chart)
- Average Renewable Energy Trend by Month(line chart)
- Renewable Energy Trend by Year(line chart)



## Dashboard 2: State and Regional Analysis

### KPI Cards

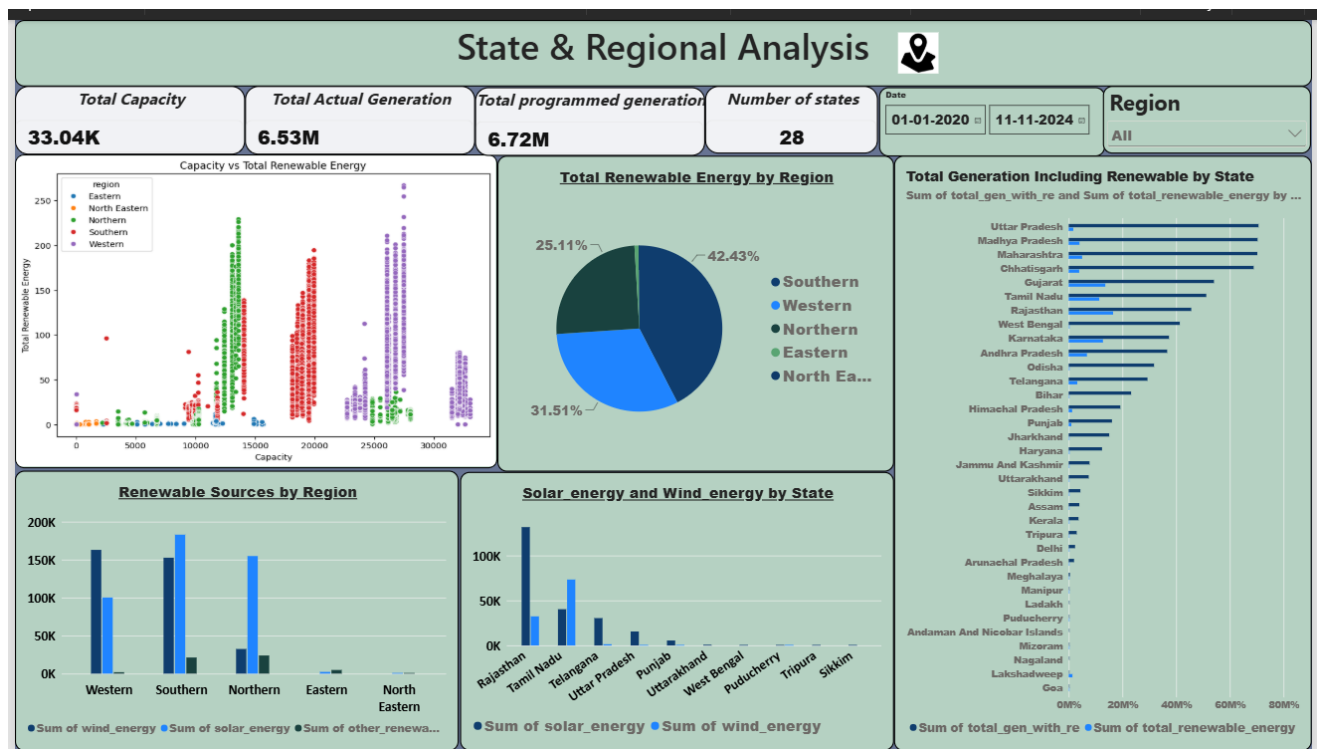
- Total capacity
- Total Actual Generation
- Total Programmed Generation
- Number of states

### Charts

- Capacity vs total renewable energy(scatter plot)
- Total Renewable energy by region(Pie chart)
- Renewable sources by region(column chart)
- Solar and wind energy by state(column chart)
- Total energy generation including renewable by state(Bar chart)

### Interactive Features

- Date Slicer
- Region Slicer



## 9. Insights :

- Solar energy accounts for the largest share of total renewable generation (441.31K) out of (841.30K) followed by wind energy (348.12K).
- Rajasthan (165K), Gujarat (136K), Karnataka (127K), and Tamil Nadu (114K) lead in renewable energy production.
- The Southern (42.43%) and Western (31.51%) regions contribute the bulk of total renewable energy generation.
- Total production grew continuously from 2020 through 2023, peaking around 200K before experiencing a slight dip in 2024.
- Strong solar resource availability drives Rajasthan's top ranking, whereas wind capacity is concentrated heavily in states like Tamil Nadu and the Western/Southern regions.
- Seasonal dynamics impact output: wind production peaks significantly during mid-year months (June–August), while solar output remains relatively steady year-round.
- Actual generation (6.53M) closely tracks total programmed generation (6.72M), indicating reliable scheduling and grid coordination.
- High-performing states (Rajasthan, Gujarat, Karnataka, Tamil Nadu) will continue to drive national renewable expansion.
- Mid-year periods will consistently experience generation spikes driven by seasonal surges in wind availability.
- The renewable portion of total energy generation is expected to expand further as total installation capacity increases.
- In underperforming states, efforts should be made to improve renewable energy productions.

## **10.Conclusions:**

- The analysis shows clear variations in renewable energy generation across states, regions, and time periods, with solar and wind contributing significantly to the renewable energy mix.
- The study highlights seasonal and regional patterns in renewable generation, helping identify areas with higher renewable-energy potential and generation performance.
- Overall, the findings provide useful insights for monitoring renewable energy performance, identifying growth opportunities, and supporting data-driven energy planning.